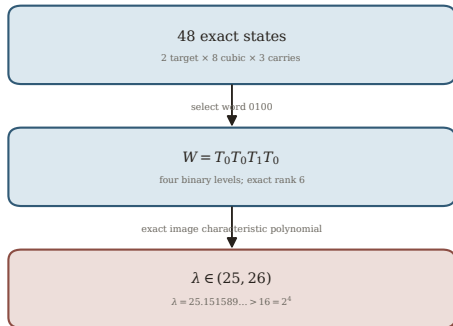


A zero-time digit cycle exceeds the factor-two threshold

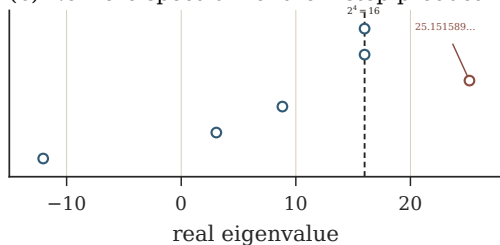
Exact integer transfer · word 0100 (least-significant bit first) · integrated heat-weighted estimate remains open

(a) Exact cycle reduction

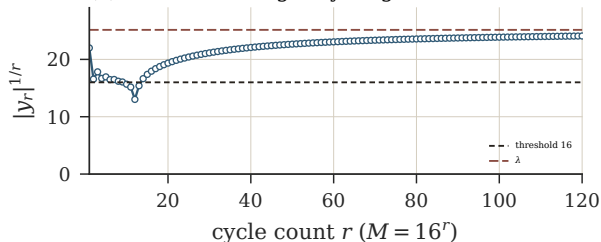


Consequence: no common full-state norm with $\|T_0\|, \|T_1\| \leq 2$ at zero heat time

(b) Nonzero spectrum of the 4-step product



(c) Reachable target-cycle growth



Interpretation: the supercritical mode is exact and reachable, so pointwise full-state contraction is impossible. Heat weighting and simplex integration must be retained before taking a norm.